

Volume 53

Engineering Standards, Testing & Quality Assurance

Status: Core Engineering Governance

Vision

Project Sentinel shall be built with the same discipline expected from enterprise infrastructure software.

Every module must be:

Documented

No exceptions.

Quality Philosophy

Quality is not a release phase. Quality is the architecture.

Every feature must be designed, implemented, tested, reviewed, and verified before release.

Engineering Standards

All Python code must follow:

Clear package boundaries

Definition of Done

A module is not complete until it has:

Architecture specification

Testing Layers

Project Sentinel uses multiple testing layers:

Unit Tests

Test Traceability

Every test links to:

Requirement ID

This gives full traceability from design to code.

Security Testing

Security testing includes:

Authentication tests

Accessibility Testing

Because WCAG is mandatory, every UI component must pass:

Keyboard navigation

Performance Testing

Performance tests measure:

Event throughput

Regression Testing

No change should silently break:

Object lifecycle

Release Gates

A release cannot ship unless:

All critical tests pass

NEW CORE ENGINE

Quality Gate Engine

Automatically evaluates whether a module or release is ready.

NEW CORE ENGINE

Test Orchestration Engine

Runs tests across:

Core kernel

NEW CORE ENGINE

Architecture Compliance Engine

Checks whether the implementation still follows the blueprint.

Examples:

No forbidden dependencies

PS-ADR-0048

Quality as Architecture

Status: Accepted

Decision

Project Sentinel shall treat engineering quality, testing, documentation, accessibility, and security validation as architectural requirements rather than optional release activities.

Rationale

Long-lived cybersecurity platforms require disciplined engineering to remain trustworthy, maintainable, and secure over time.

Strategic Principle

If it cannot be tested, it cannot be trusted.

Next: **Volume 54 - Enterprise Scaling, Resilience & Performance Architecture.**

=====[USER]=====

Good and continue

=====[ASSISTANT]=====

Excellent.

PROJECT SENTINEL BLUEPRINT

Volume 54